

3.Implement MINIMAX Algorithm

Aim :

To Use the MINIMAX algorithm to evaluate the optimal move in a game

Procedure:

1. Define a list of terminal values (leaf nodes).
2. Apply recursive MINIMAX from root to leaves.
3. Alternate between max and min at each level.

Python Code:

```
def minimax(depth,nodeIndex,isMax,scores,h):
    if depth==h:
        return scores[nodeIndex]
    if isMax:
        return max(minimax(depth+1,nodeIndex*2,False,scores,h),
                    minimax(depth+1,nodeIndex*2+1,False,scores,h))
    else:
        return min(minimax(depth+1,nodeIndex*2,True,scores,h),
                    minimax(depth+1,nodeIndex*2+1,True,scores,h))
    scores=[3,5,6,9,1,2,0,-1]
h=3
print("Optimal value:",minimax(0,0,True,scores,h))
```

Output:

Optimal value: 5

Result :

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.

